Roll No. .....

MAB-101
Examination –Dec- 2022
B.Tech. I Sem : Common for all Branches
Linear Algebra and Calculus

Time : 2 Hrs

Max. Marks : 60
Min. Marks : 19

Note: Total number of questions are 05. All Questions are compulsory. Each Question has 4 parts (a, b, c, d). Part a & b are compulsory while Part c & d has internal Choice. Assume missing data, if any.
Word limit be observed as follows:
Part a – Max 50 words, Part b – Max 50 words,
Part c – Max 100 words and Part d – Max 400 words.
Word limit NOT to be followed for diagram, numerical, derivation.

Q.1 (a) Write the statement of Maclaurin's theorem Theorem. 02

(b) Expand **logx** in power of **(x – 1)** by taylor's theorem. 02

(c) Find the first four terms in the expansion of **log(1 + sin x)** by Maclaurin's theorem 03

OR

Find the radius of curvature of the following curves at the points indicated against them 03
$\sqrt{x} + \sqrt{y} = \sqrt{a}$, (1/4, 1/4).

(d) Discuss the maxima and minima of the function **U = Sinx Siny Sin (x+y).** 05

OR

Find the radius of curuature at the Pamt **"t"** of the curve. 05
x = 3a cost – a cos3t, y = 3a sint- a sin3t

Q.2 (a) If $u = \sin^{-1}(x/y) + \tan^{-1}(y/x)$ Show that $x\,\partial u/\partial x + y\,\partial u/\partial y = 0$ 02

(b) State and prove Euler's theorem 02

(c) if $u = (x^2 + y^2 + z^2)^{-1/2}$ Show that $\partial u/\partial x + \partial u/\partial y + \partial u/\partial z = -u$ 03

OR

If $x^x y^y z^z = c$ then show that $\delta^2 z/\delta x \delta y = -(x \log ex)^{-1}$. 03

(d) (d) If $u = f(r)$, Where $r^2 = x^2 + y^2$ Show that $\partial^2 u/\partial x^2 + \partial^2 u/\partial y^2 = f''(r) + 1/r\, f'(r)$. 05

OR

If $u = \sin^{-1}\left(\dfrac{x^{1/3} + y^{1/3}}{x^{1/2} + y^{1/2}}\right)^{1/2}$ then show that 05

$$x^2\frac{\partial^2 u}{\partial x^2} + y^2\frac{\partial^2 u}{\partial y^2} + 2xy\frac{\partial^2 u}{\partial x \partial y} = \frac{\tan u}{144}(13 + \tan^2 u)$$

Q.3 (a) Evaluate 02

$$\int_0^2 \int_0^1 (x^2+y^2)\, dx\, dy$$

02

(b) Find the limit as $n \to \infty$ of the series

$$\frac{1}{n} + \frac{n^2}{(n+1)^2} + \frac{n^2}{(n+1)^3} \ldots\ldots + \frac{1}{8n}$$

03

(c) Find the area include between the parabola $y^2 = 4ax$ and $x^2 = 4ay$

OR

03

Find the limit as $n \to \infty$ of the series

$$\left[\left(1+\frac{1}{n^2}\right)\left(1+\frac{2^2}{n^2}\right)\left(1+\frac{3^2}{n^2}\right)\ldots\ldots\left(1+\frac{n^2}{n^2}\right)\right]^{\frac{1}{n}}$$

(d) Evaluate $\int_0^4 \int_0^x \int_0^{x+y} z\,dz\,dy\,dx$ 05

OR

Change the order of integration in $\int_0^1 \int_{x^2}^{2-x} xy\,dx\,dy$ and hence evaluate it. 05

Q.4 (a) Find the characteristic root's of the matrix $A=\begin{bmatrix} 5 & 4 \\ 1 & 2 \end{bmatrix}$ 02

(b) Define Echelon form of a matrix with example 02

(c) Reduce the Matrix to normal form and find it`s rank. 03

$$A = \begin{bmatrix} 8 & 1 & 3 & 6 \\ 0 & 3 & 2 & 2 \\ -8 & 1 & -3 & 4 \end{bmatrix}$$

OR

Verify Calley – Hamilton theorem for matrix. 03

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 5 \\ 3 & 5 & 6 \end{bmatrix}$$

(d) Check the consistency of the following system of equations. 05

$$7x_1+2x_2+3x_3 = 16$$
$$2x_1+11x_2+5x_3 = 25$$
$$x_1+3x_2+4x_3 = 13$$

OR

Determine the eigen values and eigen vectors of the matrix

$$A=\begin{bmatrix} 6 & -2 & 2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{bmatrix}$$

(a) Define Equivalent, Contradiction.

(b) If p and q are two statements prove that $(p \wedge q) \Rightarrow p$ is a tautology.

(c) State and prove Demorgan;s law.

OR

(1) Degree of vertex (2) Walk (3) Isomorphic graph.

(d) Express the following function into disjunctive normal form:

$$F(x, y, z) = (x+y)(x+z')+(y+z')$$

OR

Simplify the following $(a+b).(a'+b).).(a'+b').=a'b'$

$$pqr+pqr'+pq'r+p'qr = pq+qr+rp$$

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